

Crop Management in Organic and Biological Agriculture

Land Sciences

May, 2026

Crop Management in Organic and Biological Agriculture (A32362)

Short Title:	Organic Crop Management	
Faculty	Science and Computing	
Department	Land Sciences	
Cluster	Agriculture	
Author	JGERAGHTY	
Credits	10	Level: Postgraduate

Description of Module / Aims

The module content will review crop production techniques that maintain and enhance soil organic matter and lead to carbon sequestration including crop associations and rotations. Conversion to organic crop production will be described and discussed. Plant nutrition requirements will be studied including the role of macronutrients and micronutrients to improve plant health and resilience to different diseases and pests. IPM strategies to enhance sustainable crop production and profitability will be presented and evaluated. The role of new and emerging gene editing technologies will be reviewed and their relevance to organic and biological agriculture appraised. The overarching objective of this course is for graduates to understand the inter-relationship between achieving balanced plant development nutrition for climate resilient and profitable crop production.

Programmes

SCIE-0081	MSc in Organic and Biological Agriculture (WD_SOBIA_R)	1 / 1 / M
SCIE-0081	Postgraduate Programmes in Science (WD_SGRAD_XX)	5 / 0 / E

Indicative Content

- Crop and forage establishment and management practices to maintain soil health e.g. conservation agriculture
- Managing the transition from conventional practice to regenerative agriculture and organic and biological systems of production
- Beneficial crop and plant interactions -- crop associations, rotations, combi-cropping, inter-cropping and relay cropping
- Integrating diverse cover crops and maintaining crop residue covers to improve nutrient cycling and recycling in soils
- Plant nutrition -- macro and micro nutrient interactions necessary for optimum plant health status
- Integrated pest management strategies to maintain plant health and resilience to pests and diseases
- Sustainable use of bio-pesticides, biotechnology, novel crops, and plant protection products for sustainable crop production
- Policy and regulation related to organic crop production in the F2F Strategy, the CAP, and the Programme for Climate Action

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate and explain a detailed knowledge of key field management practices in sustainable cropping practices.
2. Describe and prepare beneficial crop and plant associations, interactions, and rotations for enhanced output and production.
3. Combine different cover crop species and employ plant root interactions to enhance soil biological, chemical and physical processes.
4. Assess and evaluate the role of macro and micro nutrients from inorganic and organic sources for healthy crop development.
5. Analyse and select practices and techniques to improve plant health status and develop biological pest and disease management.
6. Critically appraise and evaluate the scientific evidence for the application of gene-edited, modified and novel crop technologies at farm level.
7. Assemble, assess and interpret crop research and field trial study findings and analyse and review their application and appropriateness for in-field practice.

Learning and Teaching Methods

- Online lectures, guest lectures, laboratory practical, podcasts
- Field visit and assessment for class discussion
- Self-directed case-study for crop or forage production

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4,5,6,7
Continuous Assessment	50%	1,2,3,4,5,7
<i>Case Studies</i>	30%	1,2,3,4,5,7
<i>Assignment</i>	20%	1,2,3,4,5

Assessment Criteria

- <40%:** Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.
- 40%--59%:** The learner will be able to demonstrate a good understanding of sustainable crop management techniques and production of healthy nutritious plants. The student will be able to write and present a topical case study.
- 60%--69%:** In addition to the above, the student will demonstrate more detailed knowledge of all the module topics covered, including advances crop production systems and plant nutrition. The student will also be able to write and present a good quality case study.
- 70%--100%:** The student will be able to demonstrate comprehensive knowledge and understanding of all the learning outcomes, including the vital role of plant nutrition in sustainable crop production and profitability. The student will also be able to write and present a high quality case study.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial		12
Practical		12
Online Delivery		24
Independent Learning		222

Essential Material

- Beton Jones, J. _Plant Nutrition and Soil Fertility Manual_. 2nd ed. USA: CRC Press, 2012.
- Marschner, P., ed. _Mineral Nutrition of Higher Plants_. Netherlands: Elsevier Ltd., 2012.

Supplementary Material

- Boys, E., ed. _Encyclopaedia of Pests and Natural Enemies in Field Crops_. UK: AHDB, 2014.
- Lazor , J. _The Organic Grain Grower_. 1st ed.. USA: Chelsea Green Publishing, 2013.

Requested Resources

- Room Type: Biology Lab